Biodiversity and Horticulture

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Biodiversity and Horticulture (A08344)

Short Title:	Biodiversity and Horticulture		
Faculty	Science and Computing		
Department	Science		
Cluster	Horticulture		
Author	DFLANAGAN		
Credits	5	**Level:**	Intermediate

Description of Module / Aims

This module introduces the student to the principles of ecology and ethics of biodiversity. Students will gain an understanding of the role that horticulture can play in the conservation and development of biodiversity.

Programmes

`HORT-0030`	BSc in Horticulture (`WD_SHORT_D`)	2 / 3 / M
`HORT-0030`	BSc in Horticulture (`WD_SHORT_DX`)	2 / 3 / M

Indicative Content

- Ethics of conservation and biodiversity
- Principles of ecology - biodiversity, habitats, niche, species interactions, succession, climax community, energy flow
- Range and extent of Irish Flora and key Irish habitats, horticulture and invasive species, alien species, biosafety
- Legal framework of protection of habitats and biodiversity
- Environmental impact assessment (EIS), environmental monitoring methods
- Horticultural applications to protect, enhance or re-establish biodiversity and reduce negative environmental impacts in a range of sites including commercial horticulture and amenity e.g. brown field, roads, orchards, use of reed beds, flood amelioration techniques, tree and plant selections for urban areas, land stabilization
- Importance and protection of surface water and ground water e.g. sustainable urban drainage, water framework directive
- Urban ecology: Back gardens, urban forestry, green roof, green walls, water courses, parks
- Pollinating insects and the role of horticulture in providing protection

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Propose Horticultural interventions that would benefit Biodiversity.
2. Relate the relevant national and European regulations and laws to their role in protecting biodiversity.
3. Debate the importance of conserving biodiversity and the threats facing it.
4. Distinguish between and identify native and alien species of flora and fauna.
5. Prepare a limited EIS (or a biodiversity survey) for a designated site.

Learning and Teaching Methods

- Lectures
- Practical demonstration
- Practice

- Site visit
- Site survey
- Group work activity to encourage the development of collaborative skills

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1, 2, 3
Continuous Assessment	50%	
In-Class Assessment	10%	4
Group Project	40%	5

Assessment Criteria

<40%: Does not perform biodiversity survey or practicals assessments to reasonable standard. Limited knowledge and understanding of the application of horticulture to preserve or develop biodiversity.

40%-49%: Demonstrates a limited knowledge of the subject matter. Is able to complete a biodiversity survey at a basic level. Demonstrates limited understanding in practicals.

50%-59%: Demonstrates satisfactory general knowledge of the main issues within the subject. Practicals and biodiversity survey; demonstrates a clear understanding and grasp of the subject material.

60%-69%: Demonstrates sound knowledge of the subject matter. Shows ability to analyse and logically argue in an effective and mature style. Demonstrates initiative and the ability to criticise methods used. Practicals and biological survey; demonstrates sound understanding of theory and application.

70%-100%: Demonstrates authoritative handling of complex material and a highly developed and extensive understanding of the subject. Practicals and biological survey demonstrate excellent understanding of theory. Demonstrates reading outside of essential course material.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Practical	12	
Independent Learning	87	

Essential Material

- Fossitt, J. _A guide to Habitats in Ireland_. Dublin: The Heritage Council, 2000.

Supplementary Material

- Douglas, I. and P. James. _Urban ecology_. Devon: Taylor & Francis, 2015.

- Dunnett, N. and N. Kingsbury. _Planting Green Roofs and Living Walls_. Sheffield: Timber Press, 2008.